

HANYANG FRANK LIU

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EDUCATION

Columbia University

New York, NY

B.S. in Computer Science | GPA: 4.114/4.0

Expected May 2029

Courses: Data Structures in Java (A+), Advanced Programming (**C-language**, A), Multivar Calculus (A+), Linear Algebra (A), Probability for Engineers (A+)

Awards: Dean's List Fall 2025/Spring 2026, National Merit Johnson & Johnson Scholarship

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, C, HTML/CSS, MatLab, JS/Javascript

Frameworks/Tools: FastAPI, AWS/Amazon Web Services, Flask, Next.JS, React, Unity, Godot, Qt, Git, Matplotlib

RELEVANT EXPERIENCE

Summer@SEAS Research Fellow

May 2026 – Present

Computer-Enabled Abilities Lab at Columbia University

New York, NY

- Collaborated in fast-paced environments via the **Agile** framework to create an educational roguelike game
- Researched methods from 3 User Studies to help blind and low vision users enjoy art via **OpenCV** and **Unity VR**.

Backend Software Engineer Internship

Feb 2026 – May 2026

FurnishUp, Columbia Build Lab

New York, NY

- Designing algorithms in **Python** to optimize space efficiency, speeding up certain segments of code up to 10x
- Assessing codebase quality with **Pytest** and **Docker** to ensure readability and easier collaboration

Game Development Internship

Jul 2024 – Aug 2025

Legato, Inc.

Virtual

- Developed video game courses to improve music education using **Unity**, **C#**, and Pixilart
- Piloted product twice to groups of ~10 students and connected with diverse professionals to analyze UX

PROJECTS

Morgan Stanley Code to Give Hackathon | Python, FastAPI, AWS Bedrock, LangGraph

Mar 16, 2026

- Built nonprofit's website for campaign management using **Python+FastAPI** and won 2nd place with team of 10
- Engineered backend AI agent using **AWS Bedrock** and **LangGraph** to address key pain point for volunteers

Columbia Engineering x Amazon Bedrock Innovation Challenge | Python, AWS

Nov 7, 2025

- Applied the **AWS Bedrock** agent to create an AI chatbot that helps understand/predict hospital bills
- Boosted performance with **Python** and **AWS Lambda** to prevent hallucinations and won **Top 5 Finalist**

M3 National Math Modeling Challenge | Python, Matplotlib, LaTeX

Mar 2025

- Co-wrote **top 20/794** winning paper on addressing heat waves in England using **LaTeX** and **Overleaf**
- Coded Monte-Carlo simulations and Voronoi diagrams with K-Means clustering using **Python** and **Matplotlib**

LEADERSHIP EXPERIENCE

ADI Committee Member

Sep 2025 – Present

Application Development Initiative Club @ Columbia University

New York, NY

- Administered CodeCollab (18 teams of 3-5 people) with team of 5 to run educative coding projects
- Executed Mentorship initiative (67 participants) with another team of 5 to foster professional relationships

Event Management Committee Member

Sep 2025 – Present

Columbia Gaming Game Development

New York, NY

- Managed Game Incubator work sessions and workshopping events to over 10 teams of 3-4 people each

ACHIEVEMENTS

AIME (American Invitational Mathematics Exam) | 3x Qualifier

2023 – 2025

National 4th Place, State 2nd | FBLA Computer Problem Solving

2024

Interests | Clarinet, Band, Wushu, Undertale/Deltarune, Pickleball